

Temporal Regularized Learning: Deep Local Soft-SFA with Constructed Temporal Supervision

Anonymous authors

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Abstract

Temporal coherence is an attractive learning signal when credit assignment must remain within a layer, but slowness alone admits a collapsed solution. We introduce Temporal Regularized Learning (TEREL), a deep local rule that combines temporal coherence, gated variance expansion, and a same-layer decorrelation signal. The objective is a soft neural form of Slow Feature Analysis (SFA), viewed through the variance-invariance-covariance structure of modern regularized self-supervision. We define two roles in the method family: canonical TEREL is greedy and layer-local while retaining gradients between adjacent activations inside a short chunk; TEREL-S detaches the temporal reference and supports bounded-state samplewise execution. We derive both gradients, specify update order and boundary handling, and account separately for parameters, optimizer state, dynamic state, and batch storage. All headline experiments follow a validation-selected protocol fixed before evaluation. On label-ordered MNIST, canonical TEREL reaches $97.30 \pm 0.07\%$ test accuracy across five seeds, versus $98.34 \pm 0.08\%$ for data-presentation-matched backpropagation and $95.35 \pm 0.19\%$ for a normalization-matched random encoder. Restricting the readout to the last layer retains $97.14 \pm 0.10\%$, and TEREL-S reaches $96.29 \pm 0.07\%$. A matched label-aware Local SupCon learner reaches $96.98 \pm 0.10\%$; the paired TEREL advantage is 0.32 points. Validation controls isolate the mechanism: removing temporal coherence loses 8.18 points, removing variance expansion collapses feature scale, and removing decorrelation collapses effective rank; the lagged lateral direction has 0.76 mean cosine alignment with its same-batch target. A separately tuned, strictly samplewise TEREL-S configuration reaches $95.14 \pm 0.04\%$ validation accuracy with no retained temporal graph and fixed state. A chronological PAMAP2 stress test remains inconclusive and is reported as a secondary boundary, not the paper’s center. Together, these results establish a competitive empirical finding in the constructed MNIST regime while making precise which locality, streaming, and natural-order conclusions the evidence supports.

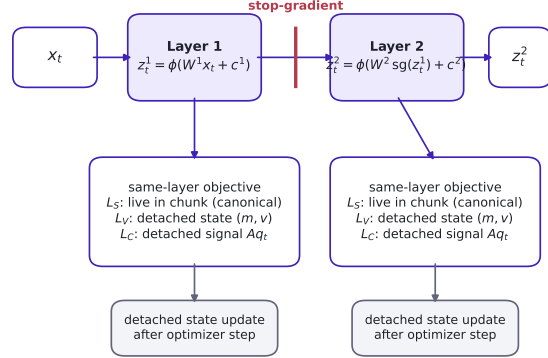
1 Introduction

Many observations change faster than the latent causes that generate them. Object identity persists while viewpoint and illumination vary; a person’s activity persists across noisy wearable-sensor readings; and the state of a dynamical system generally evolves more slowly than its measurement noise. Temporal proximity can therefore replace hand-designed paired views as a learning signal. The difficulty is that making adjacent representations equal also makes a constant representation optimal.

Classical Slow Feature Analysis (SFA) resolves this ambiguity by minimizing temporal derivatives subject to zero-mean, unit-variance, and decorrelation constraints Wiskott & Sejnowski (2002). Modern regularized self-supervised objectives arrive at a closely related structure: an invariance term is paired with variance and covariance regularizers to prevent collapse Bardes et al. (2022); Zbontar et al. (2021). The shared structure is not itself new. Our question is instead an implementation question: can this soft-SFA objective train each nonlinear layer without gradients from later layers or through temporal history?

A Local credit assignment

Every loss updates one nonlinear layer; graph cuts are explicit.

TeReL-S detaches p_t : $D^2 + 4D + 1$ state elements; no temporal graph is retained.**B Two locality regimes, retained accuracy**

Matched readouts compare methods; the samplewise variant isolates streaming.

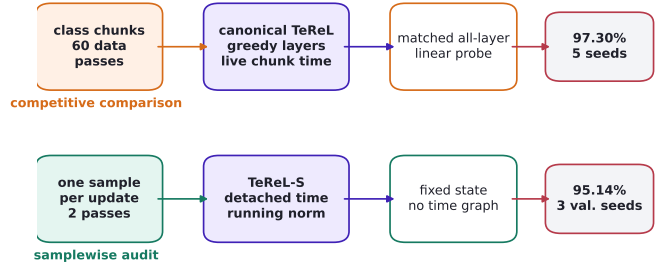


Figure 1: **A:** Each TEREL layer receives temporal-coherence, gated-variance, and lateral-decorrelation signals without an error path from later layers. **B:** Canonical TEREL retains short within-chunk temporal gradients; TEREL-S detaches them and admits bounded-state samplewise execution. Experiments use validation-only selection, matched probes, repeated seeds, and explicit compute accounting.

TEREL answers with three same-layer signals. A temporal term attracts the current activation to its preceding activation. A neuron-local term expands activity while its detached running variance lies below a target. A dense same-layer operator tracks off-diagonal second moments and supplies a detached approximation to the covariance-penalty gradient. Detaching the input to every subsequent layer makes the parameter update spatially local in the credit-assignment sense. Canonical TEREL retains the preceding activation in a short within-chunk graph and is therefore layer-local but not strictly online. TEREL-S detaches that reference, making the rule causal and bounded-history. Minibatches may still execute either variant efficiently; batching is not itself the locality claim.

This distinction matters because three claims are easily conflated. First, local credit assignment does not imply sparse communication: the lateral operator communicates within a layer. Second, bounded history does not imply constant state in width: a dense width- D lateral operator costs D^2 . Third, an objective without label terms is not self-supervised if labels were used to construct its temporal order. We therefore use label-ordered MNIST as a controlled temporal-supervision study. Chronological wearable streams with activity classes withheld from the encoder loss provide a secondary natural-order stress test; their inconclusive outcome is not allowed to displace the controlled competitive result.

The paper makes five contributions:

1. We formulate TEREL as a deep, layer-local soft-SFA rule and locate its contribution relative to SFA, Incremental SFA, Bio-SFA, VICReg, and gradient-isolated predictive, contrastive, and blockwise self-supervised learning.
2. We give the implemented objective, exact activation and local parameter gradients for detached and undetached time, including the canonical within-layer normalization Jacobian, the detached state updates, and a quantitative error decomposition for the shifted lateral heuristic, and measure the successful lagged operator against its same-batch direction.
3. We demonstrate competitive constructed-MNIST performance: canonical TEREL reaches 97.30%, versus 98.34% for backpropagation with the same data presentations and 95.35% for a normalization-matched random encoder. Matched Local SupCon and direct-covariance controls bound the attribution to TEREL’s soft-SFA construction.

4. We demonstrate tuned batch-size-one learning with fixed state and no temporal graph, and report parameters, optimizer state, dynamic state, peak memory, compute proxies, and throughput separately.
5. We provide validation-only selection, matched architectures and readouts, five evaluation seeds, raw results, uncertainty, and artifact-level provenance.

2 Temporal Regularized Learning

2.1 From constrained SFA to a soft local objective

Let x_t be a temporal stream and let layer ℓ produce, for local minibatch $\mathcal{B}$,

$$a_t^\ell = W^\ell z_t^{\ell-1} + c^\ell, \quad h_{\mathcal{B}}^\ell = N_{\eta^\ell, \mathcal{B}}(a_{\mathcal{B}}^\ell), \quad z_t^\ell = \phi(h_t^\ell). \quad (1)$$

Here N is either the identity or the within-layer normalizer with affine parameters η^ℓ . The canonical encoder uses training-mode BatchNorm, so $N_{\eta, \mathcal{B}}$ couples examples inside the current local minibatch; it does not introduce an error path from a later layer. SFA searches for features that change slowly while excluding constant and duplicated solutions. In one common statement it minimizes $\mathbb{E}\|z_t - z_{t-1}\|_2^2$ subject to zero mean, unit variance, and decorrelated outputs Wiskott & Sejnowski (2002). TEREL replaces hard population constraints by detached running state and local soft terms. This is a particular neural parameterization and optimizer, not a new discovery of the SFA objective.

For one layer, consider B activations $z_t \in \mathbb{R}^D$. A Boolean boundary b_t marks the start of a subject, recording, or constructed chunk. Let $I_t = 1 - b_t$, additionally set to zero when no stored predecessor exists, and let $N_I = \sum_t I_t$. For minibatch k , the state available before the step is the detached triple $(m_{k-1}, v_{k-1}, A_{k-1})$; we suppress its batch index in the local objective:

$$L_S = \frac{1}{N_I D} \sum_{t: I_t=1} \|z_t - p_t\|_2^2, \quad (2)$$

$$L_V = -\frac{1}{BD} \sum_{t=1}^B \sum_{j=1}^D [\gamma - v_j]_+ (z_{tj} - m_j)^2, \quad (3)$$

$$L_C = \frac{1}{BD} \sum_{t=1}^B (z_t - m)^\top A q_t, \quad (4)$$

$$L = \lambda_S L_S + \lambda_V L_V + \lambda_C L_C. \quad (5)$$

If $N_I = 0$, L_S and its gradient are defined as zero. In canonical TEREL, $p_t = z_{t-1}$ inside a short chunk and $q_t = \text{sg}(z_t - m)$. The input to each subsequent layer remains detached, so these within-chunk temporal gradients never become gradients across depth. In TEREL-S, $p_t = \text{sg}(z_{t-1})$ (or the detached last activation from the preceding call). Boundary flags invalidate temporal pairs in both cases.

The minus sign in Eq. (3) is essential. With the gate $g_j = [\gamma - v_j]_+$ treated as state, minimizing L_V pushes a currently active unit away from its mean until its running variance reaches γ . The term is a local gradient surrogate for a variance constraint; it is not the usual nonnegative VICReg scalar variance penalty. Its effectiveness depends on the statistics timescale and normalization. We therefore treat state rates, normalization, and activation choice as algorithmic hyperparameters and select them on validation data.

2.2 Lateral decorrelation as a gradient proxy

Define the detached same-minibatch off-diagonal second-moment operator

$$C_k = \text{offdiag} \left(\frac{1}{B} \sum_t (z_{k,t} - m_{k-1})(z_{k,t} - m_{k-1})^\top \right). \quad (6)$$

The direct control used in our ablations minimizes $L_{\text{direct}} = \|C_k\|_F^2/D$. With m_{k-1} detached, its activation gradient is $4C_k(z_{k,t} - m_{k-1})/(BD)$. TEREL instead uses the operator A_{k-1} available before the step and Eq. (4); because q_t and A are detached, the encoder receives $A_{k-1}q_{k,t}/(BD)$. If $A_{k-1} \approx C_k$ and $q_{k,t} = z_{k,t} - m_{k-1}$, this matches the direct gradient direction up to a positive factor four. It is a lagged proxy for the gradient, not an equality between scalar objectives.

The shifted heuristic uses the preceding centered activation as q_t . Let $u_t = z_t - m$. Its error relative to the direct signal obeys

$$\|Aq_t - Cu_t\|_2 \leq \|A - C\|_2 \|u_t\|_2 + \|A\|_2 \|q_t - u_t\|_2. \quad (7)$$

The two terms expose separate failure modes: an inaccurate lateral estimate and a representation that changes too quickly for a one-step shift. The bound does not guarantee a useful shifted signal, so we report this variant only as an ablation.

2.3 Exact local gradients

Write $g = [\gamma - v]_+$. For TEREL-S the loss gradient with respect to the post-activation is

$$e_t = \frac{2\lambda_S I_t}{N_I D} (z_t - p_t) - \frac{2\lambda_V}{BD} g \odot (z_t - m) + \frac{\lambda_C}{BD} Aq_t. \quad (8)$$

Let $r_s = e_s \odot \phi'(h_s^\ell)$ and let $J_{st}^N = \partial h_s^\ell / \partial a_t^\ell$ be the minibatch Jacobian block of the within-layer normalizer. The input $z_t^{\ell-1}$ to each learned layer is detached. The exact same-layer chain rule is therefore

$$u_t = \sum_{s \in \mathcal{B}} (J_{st}^N)^\top r_s, \quad \frac{\partial L^\ell}{\partial W^\ell} = \sum_t u_t (z_t^{\ell-1})^\top, \quad \frac{\partial L^\ell}{\partial c^\ell} = \sum_t u_t, \quad \frac{\partial L^\ell}{\partial \eta^\ell} = \sum_s \left(\frac{\partial h_s^\ell}{\partial \eta^\ell} \right)^\top r_s. \quad (9)$$

For $N = \text{Id}$, $J_{st}^N = \delta_{st} I$ and the weight update reduces to the presynaptic outer product $\sum_t (e_t \odot \phi'(a_t^\ell))(z_t^{\ell-1})^\top$. Training-mode BatchNorm instead mixes the output-side factors across examples in $\mathcal{B}$ and also updates η^ℓ . Autograd evaluates this local Jacobian exactly. In both cases no later-layer error enters Eq. (9); the simple per-example factorization applies only to the identity-normalizer case.

Canonical TEREL leaves adjacent $p_t = z_{t-1}$ inside the short chunk graph. Its temporal contribution is

$$\frac{2\lambda_S}{N_I D} [I_t(z_t - z_{t-1}) - I_{t+1}(z_{t+1} - z_t)], \quad (10)$$

with missing and boundary-crossing terms omitted. The second term depends on a future activation within the chunk. We therefore describe canonical TEREL as layer-local but not temporally online; TEREL-S is both layer-local and temporally local. Across minibatch calls, even canonical TEREL reads its stored predecessor as detached state. In the fixed MNIST construction, a mean 151.2 of 46,866 valid edges per data presentation cross such a boundary (0.32%; 9,073 edges over the 60-pass encoder schedule); all remaining valid edges use the live within-call gradient in Eq. (10).

2.4 State update and execution order

After the optimizer step for minibatch k , state is updated under `no_grad` from the activations computed for that step:

$$m_k = \rho_m m_{k-1} + (1 - \rho_m) \bar{z}_k, \quad (11)$$

$$v_k = \rho_m v_{k-1} + (1 - \rho_m) \frac{1}{B} \sum_t (z_{k,t} - m_{k-1})^2, \quad (12)$$

$$A_k = \rho_A A_{k-1} + (1 - \rho_A) \text{offdiag} \left(\frac{1}{B} \sum_t (z_{k,t} - m_{k-1})(z_{k,t} - m_{k-1})^\top \right). \quad (13)$$

Thus $\rho_A = 0$ makes $A_k = C_k$, but the loss for batch k has already used A_{k-1} ; the successful configuration is a one-minibatch-lagged proxy. The pre-step mean m_{k-1} is used in Eqs. (12) and (13). For TEREL-S, the

final activation and its centered value are stored as detached temporal state. No activation graph survives the call.

Algorithm 1 One TEREL-S step for a deep encoder.

Require: input batch/stream segment X , boundary flags b , layer parameters $\{W^\ell, c^\ell, \eta^\ell\}$, and states $\{m^\ell, v^\ell, A^\ell, p^\ell\}$

- 1: $h \leftarrow X$
- 2: **for** layer $\ell = 1, \dots, L$ **do**
- 3: $z^\ell \leftarrow \phi(N_{\eta^\ell, \mathcal{B}}(W^\ell \text{sg}(h) + c^\ell))$
- 4: construct detached predecessors p_t^ℓ and valid pairs from b
- 5: compute L^ℓ by Eq. (5)
- 6: $h \leftarrow \text{sg}(z^\ell)$
- 7: **end for**
- 8: backpropagate $\sum_\ell L^\ell$ only within each layer, including $N_{\eta^\ell, \mathcal{B}}$; step optimizer
- 9: **for** layer $\ell = 1, \dots, L$ **do**
- 10: update detached state by Eqs. (11) to (13)
- 11: **end for**

2.5 Variants and claim boundary

Table 1: Variants used in the experiments. “Time graph” and “depth graph” indicate whether autograd crosses those boundaries.

Method	Temporal reference	Decorrelation	Graph scope
TEREL	live previous in chunk	lagged A_{k-1} , current q_t	same layer/chunk time
TEREL-S	detached previous	lagged A_{k-1} , current q_t	same layer/current step
Shifted proxy	detached previous	tracked A , previous q_t	same layer/current step
Direct control	live previous in chunk	exact in-batch $\ C_k\ _F^2$	same layer/chunk time
Local SupCon	label positives	supervised contrastive	same layer/current batch

Our primary locality claim is spatial: parameters receive no gradient from another layer. Canonical TEREL keeps short-range temporal credit within a chunk; TEREL-S additionally cuts credit across time and is online in the bounded-history sense when executed at batch size one. We do not equate these properties with biological plausibility or measured hardware efficiency.

2.6 State and memory cost

For layer width D , dynamic TEREL state contains mean D , variance D , lateral operator D^2 , previous activation D , previous centered activation D , and one validity flag: $D^2 + 4D + 1$ elements. A bounded streaming normalizer, when used, adds location and scale vectors of size $2D$. Both totals are constant in stream length and quadratic in width. Encoder parameters, optimizer moments, batch activations, and temporary matrix products are separate costs. In particular, AdamW adds approximately two parameter-sized moment buffers after its first step. The lateral state is discarded at inference, but its training cost is real. Batch-size-one execution removes batch growth and temporal graphs; it does not remove the dense lateral matrix.

3 Relation to prior work

SFA and incremental SFA. SFA is the objective-level ancestor of TEREL: temporal slowness, variance, and decorrelation are its defining ingredients Wiskott & Sejnowski (2002). Batch linear SFA whitens observations and solves a minor-component problem for their derivatives. IncSFA replaces those batch operations

with covariance-free incremental PCA and minor-component updates, supports episodic boundaries, and has Hebbian/anti-Hebbian update forms Kompella et al. (2012). It is therefore a stronger online precedent than a comparison to batch SFA alone. Our empirical baseline is a provenance-documented Python 3 port of the authors’ published linear algorithm, validated against batch SFA on a synthetic stream. TEREL differs by applying a soft objective directly to each nonlinear learned layer, at the cost of a dense tracked lateral operator.

PowerSFA instead makes approximate whitening differentiable and thereby trains deep SFA models end to end Schöler et al. (2018). This establishes that a neural, gradient-trained SFA parameterization is not itself the contribution here; TEREL removes the gradient path across learned layers and, in TEREL-S, across temporal history. Deep temporal-coherence objectives also predate this work: for example, Slow and Steady Feature Analysis regularizes first- and higher-order changes across video tuples in a convolutional encoder Jayaraman & Grauman (2016). TEREL returns to first-order adjacency to study the distinct question of layer-local soft constraints and bounded temporal state.

Bio-SFA and trace learning. Bio-SFA derives an online neural network for the SFA problem with local synaptic updates Lipshutz et al. (2020). This prevents us from presenting local online SFA as absent from prior work. TEREL instead contributes a different deep soft-penalty parameterization and an explicit stop-gradient implementation that is easy to compose with standard nonlinear layers. Földiák-style trace learning also exploits temporal persistence Földiák (1991); its positive activation trace is not algebraically the same as the signed difference in Eq. (8).

Gradient-isolated predictive and contrastive learning. Greedy InfoMax trains a stack of gradient-isolated modules with local InfoNCE objectives that predict across the natural order of images, audio, and video Löwe et al. (2019). It is therefore a direct precedent for combining temporal structure with greedy depth-local representation learning. CLAPP makes contrastive prediction causal and local in both space and time through pre/post activity, predictive dendritic input, and broadcast modulation, while still relying on positive and negative examples Illing et al. (2021). LoCo instead improves paired-view local contrastive learning with overlapping blocks, which allows the next block’s local loss to influence parameters in the overlap and therefore relaxes strict gradient isolation Xiong et al. (2020). These methods preclude claiming either local temporal prediction or gradient-isolated deep learning as new here. Relative to them, TEREL studies a noncontrastive soft-SFA loss with explicit variance and redundancy controls; TEREL-S uses no negative set or predictive decoder, but pays for a dense lagged same-layer operator. Our constructed MNIST protocol is also label-assisted through ordering, unlike their principal unlabeled settings.

Regularized and blockwise self-supervision. VICReg and Barlow Twins use variance/covariance or cross-correlation penalties to prevent collapse without negatives Bardes et al. (2022); Zbontar et al. (2021). Blockwise self-supervised learning has already shown that VICReg, Barlow Twins, and contrastive losses can train network blocks without end-to-end backpropagation Siddiqui et al. (2024). TEREL is therefore not distinguished by “a local VICReg loss” alone. Its narrower focus is temporal positives, detached streaming moments, and a same-layer covariance-gradient proxy that can execute one observation at a time.

Two boundaries sharpen that comparison. Siddiqui et al. find simultaneous block training materially stronger than sequential freezing, indicating that forward-path co-adaptation remains useful even when gradients stop at block boundaries Siddiqui et al. (2024); the canonical TEREL result instead uses greedy layerwise training. Datta et al. likewise apply VICReg losses at individual layers, using original/augmented image pairs and current-batch variance and covariance terms Datta et al. (2025). In contrast, TEREL uses a temporal relation, a gated detached variance state, and a lagged lateral gradient proxy. Spectral analyses of SSL further show that the chosen positive relation can determine downstream geometry when it aligns with task labels Balestriero & LeCun (2022). Consequently, our class-chunk MNIST result is a study of constructed pair supervision and requires a matched label-aware local baseline; it is not evidence that arbitrary natural adjacency reveals class structure.

Table 2: Conceptual comparison. Costs omit learned feedforward parameters.

Method	Temporal objective	Execution	Learned mapping	Auxiliary state
Batch SFA	derivative minimization	batch eigensystem	linear	covariance and whitening
IncSFA	derivative minimization	sample based	linear; deep cascades possible	incremental PCA and MCA state
Bio-SFA	derivative minimization	online neural dynamics	linear network in reported derivation	synaptic/dual variables
Greedy InfoMax	natural-order InfoNCE	greedy isolated modules	deep nonlinear modules	context/predictor and negatives
CLAPP	causal contrastive prediction	local in depth and time	deep nonlinear layers	predictive input, modulation, negatives
LoCo	paired-view contrastive	overlapping minibatch blocks	deep nonlinear blocks	projection heads and negatives
Blockwise SSL	paired-view invariance	mini-batch blocks	deep nonlinear blocks	batch statistics
TEREL family	temporal soft-SFA	sample or minibatch	deep nonlinear layers	$D^2 + 4D + 1$ per layer

4 Experimental protocol

The experiments answer four questions. First, how does canonical TEREL compare with data-presentation-matched backpropagation, normalization-matched and no-normalization random encoders, and a label-aware local learner under the constructed supervision? Second, is that result carried only by early layers or retained by the final layer? Third, can TEREL-S learn non-collapsed representations one sample at a time with bounded state? Fourth, as a secondary stress test, does the temporal-coherence advantage transfer to a natural wearable stream?

4.1 Data roles and leakage control

MNIST controlled temporal supervision. The official 60,000-example training split is stratified with seed 1701 into 50,000 training and 10,000 validation examples; the official 10,000-example test split is used only for final evaluation. Within each encoder epoch, training examples are randomly arranged into same-class chunks of length 16; all chunk starts are temporal boundaries. Labels create the persistent order but do not enter the TEREL loss. This controlled setting isolates whether temporal continuity can provide a useful local teaching signal; it is not presented as unlabeled self-supervision.

PAMAP2 secondary natural-order stress test. PAMAP2 contains 100-Hz wearable measurements and activity annotations from nine subjects Reiss (2012). We use its 12 protocol activities, discard transition label 0 as instructed by the dataset documentation, and downsample by a fixed stride of 10. Missing sensors are imputed with training-subject means and standardized using training-subject statistics. Subjects 1–6 train the encoder and probe, subject 7 is validation, and subject 8 is the untouched test subject. Subject 9 is excluded before model selection: its protocol recording contains only rope jumping and cannot support a 12-class test. Boundaries are inserted at subject starts, removed-label gaps, and timestamp discontinuities. The encoder iterator returns only features and boundaries, but annotation label 0 determines which transition rows are removed and where gap boundaries occur. Activity-class labels therefore do not enter the encoder loss, although annotations influence preprocessing. This experiment tests generalization of the temporal intuition; it does not replace the controlled MNIST comparison.

4.2 Models, baselines, and readouts

The primary encoder is $784 \rightarrow 512 \rightarrow 256$ with LeakyReLU (negative slope 0.01). Canonical TEREL uses affine batch normalization (update momentum 0.1), live temporal pairs within chunks, and greedy layerwise training for 30 epochs per layer. Thus both its encoder and the matched BP encoder receive 60 complete training-set presentations. Its AdamW learning rate is 3×10^{-4} , weight decay 10^{-4} , batch size 256, and

coefficients $(\lambda_S, \lambda_V, \lambda_C) = (1, 2.5, 1)$. Population momentum is 0.9; $\rho_A = 0$ writes each minibatch operator after its optimizer step, so the next minibatch uses that preceding-minibatch operator.

Backpropagation uses the same hidden dimensions, activation, normalization, optimizer, batch size, and 60 data passes, trained end to end with cross entropy. TEREL-S uses detached temporal references, joint local training, no batch normalization, $(1, 5, 1)$, and a separately validated ten-pass schedule. It is an intentionally lean temporally local variant, so its compute is reported rather than described as matched to BP. A no-normalization random reference retains the same seed-specific hidden weights but omits normalization; we keep it because it matches the no-normalization TEREL-S architecture. The normalization-matched random control instead includes the canonical affine BatchNorm modules. Its hidden weights and biases remain at initialization, as do BatchNorm scale one and offset zero. Under `no_grad`, one pass over the 50,000 training examples in stored order (batch size 256) calibrates only running means and variances with update momentum 0.1 (0.9 retention). The model then enters evaluation mode, and those fixed train-only statistics are exposed to the same all-layer probe and test examples. This is data-dependent normalization, not an encoder-parameter update.

Canonical minibatches concatenate randomly permuted class-homogeneous chunks, so each batch can mix classes while BatchNorm communicates across all 256 local examples. Under greedy training, only the active layer’s BatchNorm is in training mode: layer-one statistics are fixed before layer two is trained. Representation extraction and probing use evaluation mode with the final running statistics and learned affine parameters. Thus BatchNorm changes the within-layer information path but never carries error across depth.

Local SupCon supplies a label-aware local comparison for the constructed MNIST regime. Each layer receives a supervised-contrastive loss over every same-label positive in its minibatch; activations are detached between layers, while both layers update simultaneously. It uses the same architecture, batch normalization, batch size, 60 data presentations, all-layer probe, and test seeds as the primary comparison. A validation-only 3×2 grid varies learning rate in $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$ and temperature in $\{0.1, 0.2\}$. This six-candidate budget matches the primary method-selection budget.

The primary readout for every method is the same linear classifier over the concatenation of both hidden layers. A final-layer-only probe on the identical canonical encoder is a secondary audit of depth specialization. All probes use AdamW for 60 epochs (learning rate 0.003, weight decay 10^{-4} , batch size 2,048). This matched-access design prevents a comparison in which a local method exposes less of its representation than the baseline.

4.3 Model selection and evaluation separation

Selection uses only the 10,000-example validation split and seeds 101, 202, and 303. Candidate method configurations are compared on those seeds, after which one configuration per method is fixed for the five-seed evaluation. In particular, the Local SupCon learning-rate/temperature grid and the TEREL component and lateral-signal controls use validation data only. Reported test results are not used to choose among alternative configurations.

The streaming configuration is selected separately because batch-size-one execution changes both normalization and update count. It uses TEREL-S, bounded running normalization, two passes, learning rate 6×10^{-6} , statistics and lateral momenta 0.99, and one forward/backward step per example. No future example, retained temporal graph, or stream-length-dependent buffer is used.

4.4 Evaluation and uncertainty

Central results use seeds 1101, 1202, 1303, 1404, and 1505. We report every raw result, the mean, sample standard deviation, a small-sample Student- t interval, and a percentile 95% bootstrap interval from 10,000 resamples (analysis seed 260803). Primary contrasts pair methods by seed; Student- t intervals are primary and bootstrap intervals are retained as a sensitivity analysis. Learned-representation evidence is assessed against both the normalization-matched random encoder and the no-normalization random reference. Non-

Table 3: MNIST test accuracy (% , five seeds). All-layer readout access is matched across methods; TeReL last-layer access is a secondary audit. Values are mean $\pm$ sample SD.

Method	Accuracy
TeReL (all layers)	97.30 $\pm$ 0.07
TeReL (last layer)	97.14 $\pm$ 0.10
TeReL-S (all layers)	96.29 $\pm$ 0.07
Random, no normalization (all layers)	95.13 $\pm$ 0.27
Backpropagation (all layers)	98.34 $\pm$ 0.08

Table 4: Paired test-set accuracy differences in percentage points with 95% Student- t intervals over the five seed-paired differences.

Contrast	Difference [95% interval]
TeReL – BP	-1.04 [-1.08, -0.99]
TeReL – random, no normalization	2.17 [1.83, 2.51]
last – all layers	-0.16 [-0.37, 0.05]

collapse requires every canonical seed to have effective rank at least 20, median variance at least 0.05, and active fraction at least 0.25.

The final configurations, seeds, calibration treatment, and analysis rules are fixed before evaluation. Each selected method has one five-seed evaluation; the BatchNorm-calibrated random control uses the same seeds without a validation sweep. Mechanism and direct-covariance analyses remain validation-only and do not select replacement models. The supplement records the complete selection ledger and raw outputs.

5 Results

5.1 Competitive performance

Canonical TeReL reaches $97.30 \pm 0.07\%$, compared with $98.34 \pm 0.08\%$ for matched BP. The paired gap is 1.04 percentage points (95% paired Student- t interval [0.99, 1.08]). Relative to the BatchNorm-calibrated random encoder, TeReL gains 1.95 points ([1.64, 2.25]). The calibrated random mean is $95.35 \pm 0.19\%$, compared with $95.13 \pm 0.27\%$ for the retained no-normalization random reference; their paired difference is 0.22 points ([−0.08, 0.52]). In error-rate terms, however, its 2.70% error is 63% higher than BP’s 1.66%; the small absolute-accuracy gap should therefore not be read as error-rate parity. All five seeds satisfy the non-collapse criteria, and every intended layer and normalization group has a nonzero parameter update. The evidence therefore supports the paper’s central empirical statement within this constructed MNIST regime: a two-hidden-layer network trained by layer-local soft-SFA reaches 97.30%.

5.2 Matched label-aware local comparator

The validation-selected Local SupCon baseline reaches $96.98 \pm 0.10\%$ on the same five test seeds. Canonical TeReL is 0.32 points higher, with a paired Student- t interval of [0.13, 0.52]. Local SupCon sees every same-label positive in each minibatch, whereas TeReL sees only adjacency in the label-constructed chunks. Thus access to a generic layer-local same-class positive objective does not by itself reproduce the TeReL result in this protocol. The difference is small and benchmark-specific, so it supports a narrow attribution to the soft-SFA construction rather than general superiority over local contrastive learning.

5.3 Mechanism isolation

The validation-only one-factor audit uses the canonical protocol. Removing the temporal term loses 8.18 percentage points without collapsing rank, while fully shuffling the class-persistent order loses 7.55 points.

Table 5: Normalization-matched random-control audit on MNIST test accuracy (% , five paired seeds). The BatchNorm-calibrated encoder keeps all weights and affine normalization parameters at initialization; only train-split running statistics are updated. Values are mean $\pm$ sample SD, and contrasts use 95% Student- t intervals.

Method or contrast	Accuracy / difference
Random, no normalization	95.13 $\pm$ 0.27
BatchNorm-calibrated random	95.35 $\pm$ 0.19
TEREL	97.30 $\pm$ 0.07
TEREL – calibrated random	1.95 [1.64, 2.25]
calibrated – no-normalization random	0.22 [-0.08, 0.52]

Table 6: Matched label-aware comparison on MNIST test accuracy (% , five seeds). Local SupCon receives all same-label positives in each minibatch; both methods have layer-local credit, the same architecture, 60 data presentations, and the same all-layer probe.

Method	Accuracy
TEREL	97.30 $\pm$ 0.07
Local SupCon	96.98 $\pm$ 0.10
TEREL – Local SupCon	0.32 [0.13, 0.52]

The bracketed contrast is a paired 95% Student- t interval.

Temporal coherence is therefore the task-aligned teaching signal rather than an incidental regularizer. The two remaining terms prevent different degeneracies: without gated variance expansion, median feature variance falls to 7.7×10^{-5} ; without decorrelation, variance remains high but effective rank falls to 2.8. Batch normalization alone does not rescue either failure. The three signals thus have empirically distinct roles matching the soft-SFA interpretation.

Table 7: Validation-only mechanism audit under the canonical protocol (three paired seeds). Each intervention changes one factor. Δ is the accuracy difference from full TEREL in percentage points.

Intervention	Accuracy (%)	Δ	r_{eff}	Median var.
Full TEREL	97.28 $\pm$ 0.12	–	39.2	0.185
$\lambda_S = 0$	89.10 $\pm$ 1.57	-8.18	369.3	0.907
$\lambda_V = 0$	66.13 $\pm$ 6.32	-31.15	6.0	7.72e-05
$\lambda_C = 0$	86.44 $\pm$ 0.41	-10.84	2.8	0.916
Shuffled order	89.73 $\pm$ 0.37	-7.55	311.3	0.171

The separate lagged-operator audit covers 17,637 nonzero minibatch directions per layer across three validation seeds. The preceding-batch direction has a mean cosine of 0.76 to the same-batch direction in both layers; its relative ℓ_2 error is 0.65 and its norm is about 0.78 times as large. Thus the proxy is directionally informative but plainly not identical to a same-batch covariance gradient. Replacing it by the exact differentiable same-batch penalty, with the factor-four gradient scale matched, reaches $96.93 \pm 0.07\%$ validation accuracy versus $97.19 \pm 0.09\%$ for the audited lagged configuration. The paired direct-minus-lagged difference is -0.25 points (95% Student- t interval $[-0.42, -0.08]$). The competitive result therefore does not require an exact same-batch operator, while the measured discrepancy prevents interpreting the lagged state as one.

5.4 Depth, readout, and representation geometry

The last hidden layer retains almost all of the learned signal: its probe reaches $97.14 \pm 0.10\%$, only 0.16 points below the matched all-layer readout with a paired Student- t interval spanning zero $[-0.37, 0.05]$. Thus

Table 8: Validation audit of the lateral signal (three seeds). Alignment statistics average over seeds and all nonzero audited minibatches.

Layer	Cosine alignment	Relative ℓ_2 error	Norm ratio
1	0.757	0.651	0.785
2	0.755	0.653	0.780

the mean result is not an artifact of concatenating a strong first layer while ignoring a weak second layer, although five seeds do not resolve the small readout difference. TEREL-S reaches $96.29 \pm 0.07\%$ after only ten joint-training passes and beats its no-normalization random reference by 1.16 points ($[0.94, 1.45]$), showing that temporal detachment trades some accuracy for a stricter locality boundary rather than destroying the mechanism.

The diagnostic geometry agrees with the probe result. Canonical TEREL raises nearest-centroid accuracy from 80.21% for the no-normalization random encoder to 95.55% and the between/within-class scatter ratio from 0.24 to 0.70. Its final layer is even more class concentrated (96.22% centroid accuracy and 1.52 scatter ratio), while the all-layer representation keeps greater effective rank. This is the expected soft-SFA tradeoff: persistent factors become accessible without forcing every output direction into the same class-compressed geometry.

Table 9: MNIST representation diagnostics averaged across five seeds. B/W is between/within-class scatter, Sel. is median unit class selectivity, r_{eff} is covariance effective rank, and $|\rho_{\text{off}}|$ is mean absolute off-diagonal correlation.

Method	Centroid (%)	B/W	Sel.	r_{eff}	$ \rho_{\text{off}} $
Random, no norm. (all)	80.21	0.241	1.239	33.84	0.101
TEREL-S (all)	92.04	0.101	0.843	142.32	0.057
TEREL (all)	95.55	0.695	1.704	37.35	0.103
TEREL (last)	96.22	1.516	2.567	18.86	0.161
BP (all)	97.87	0.968	1.857	21.17	0.152

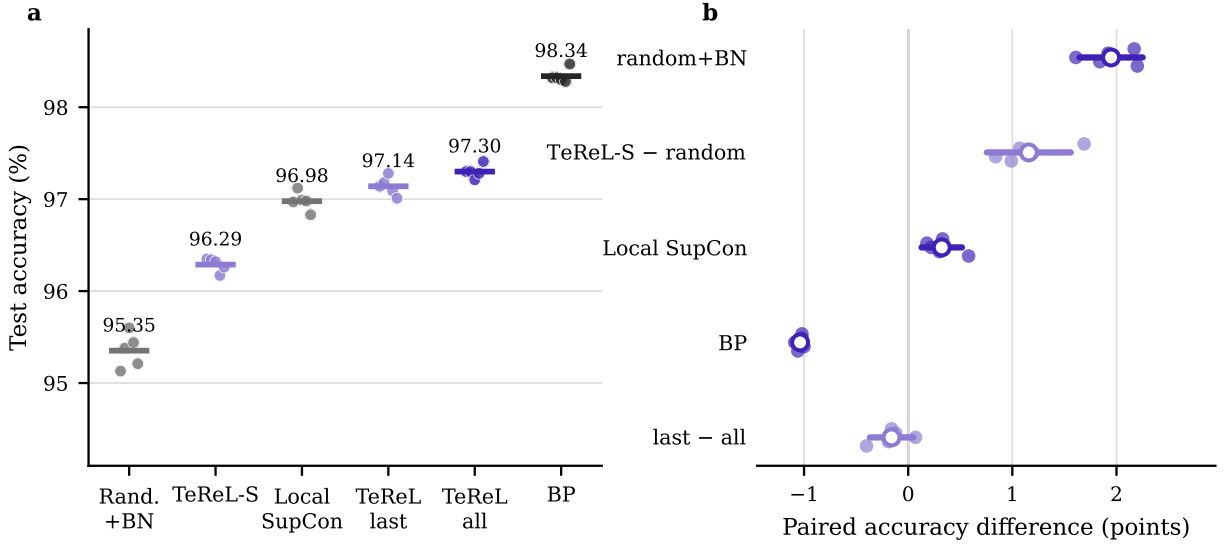


Figure 2: MNIST performance and paired contrasts. **a**: individual test seeds and means (horizontal bars). Canonical TEREL has a small absolute-accuracy gap to BP, exceeds both the normalization-matched random encoder and the matched label-aware Local SupCon learner, and retains the mean result in its final layer; temporally local TEREL-S remains above its no-normalization random reference. **b**: paired seed differences; rows are canonical TEREL minus the named comparator except for the explicitly named TEREL-S and readout contrasts. Lines are paired 95% Student- t intervals and open circles are means.

5.5 Natural temporal order on PAMAP2

The chronological/shuffled PAMAP2 contrast was introduced to answer whether the controlled temporal mechanism transfers when order is natural and labels are excluded from encoder batches. Architecture, initialization, optimizer, sample count, and probe are held fixed; only encoder order changes. The result is inconclusive: chronological and shuffled learning reach 0.307 and 0.312 macro-F1, respectively, with a paired interval spanning both directions $([-0.063, 0.053])$. All methods also exhibit a large subject-7 to subject-8 shift. We report this boundary because it is informative, but do not treat a single heterogeneous wearable split as a reversal of the controlled MNIST result or of the temporal-coherence motivation. Broader natural-stream tests need multi-subject validation and stronger shift control.

5.6 Streaming execution and cost

The samplewise configuration is selected on validation data over learning rate, state timescale, streaming normalization, accumulation, and exposure count. The selected two-pass configuration reaches $95.14 \pm 0.04\%$ accuracy across three seeds, with effective rank 112.8 (seed range 106.6–120.9) and median feature variance 0.378. It exceeds the matched no-normalization random validation mean by 0.29 points while retaining no temporal graph.

At every step, the previous activation, running moments, normalization state, and lateral matrix are detached; state size is unchanged by stream length. The serial resource seed processes one observation per optimizer step and takes 1,305 seconds, illustrating that a stock GPU/PyTorch implementation pays a large launch and update overhead. This is evidence for quality-bearing samplewise learning and bounded storage, not a claim that the present software is throughput- or energy-optimal. Encoder resource accounting for the main methods is given in Tables 12 and 14.

6 Limitations

TEREL inherits the temporal-coherence assumption: rapidly changing or aliased latent causes can make adjacent observations misleading. The MNIST ordering is synthetic and label-derived, so its high accuracy establishes the efficacy of the local temporal teaching signal rather than label-free learning. PAMAP2 is a modest, distribution-shifted sensor benchmark, and its single subject-disjoint split cannot decide the broader natural-order question. Scaling to images, video, audio, and additional natural streams remains an empirical requirement. Activity-monitoring deployments would also require consent, privacy, and surveillance-risk analysis beyond this methodological benchmark.

The dense lateral operator gives bounded history but $O(D^2)$ state and within-layer communication. Sparse or low-rank approximations need direct quality tests. The streaming result is validation evidence on MNIST, not a five-seed test-set or throughput-matched comparison; samplewise optimizer steps are slow in the present GPU implementation. Canonical TEREL also takes more wall-clock time than BP despite a lower declared operation proxy, so no hardware or energy advantage follows from these software measurements.

Canonical BatchNorm is a same-layer, cross-example communication channel and its affine parameters are learned. Depth-local credit assignment therefore does not imply a per-example local update for the headline model; the simpler outer-product factorization applies to the identity-normalizer variants.

Finally, locality here concerns credit assignment: standard optimizers, exact activation derivatives, normalization, and synchronized matrix operations remain.

7 Perspective and future directions

Natural order. SFA’s motivating regime is not class-sorted images but streams in which stable causes generate many changing observations. The PAMAP2 result shows that adjacency alone is insufficient under severe subject shift; it does not remove the underlying opportunity. Video under controlled viewpoint change, egocentric interaction, and continuous audio provide cleaner tests in which the timescale of the latent cause can be compared directly with the TEREL state and chunk timescales. A useful next study should vary those timescales rather than reduce natural order to one chronological-versus-shuffled number.

Constrained hardware. For the identity-normalizer and streaming forms, the rule’s ingredients—presynaptic activity, an output-side teaching factor, leaky state, and no error path across depth—are structurally compatible with layer-local or event-driven execution. TEREL-S further removes the temporal graph, and our samplewise experiment shows that this stricter form can retain useful accuracy. These are architectural reasons to investigate specialized implementations, not measurements of energy efficiency. Low-rank lateral state, fused state updates, and optimizer choices designed for local arrays are the concrete engineering steps needed to test that perspective.

Recurrent extension. The temporal objective also suggests a recurrent representation module whose slow state supports prediction without receiving prediction gradients. A valid recurrent comparison should isolate the representation and prediction paths, include matched random-state and BPTT controls, and repeat across seeds. We retain this as a motivated extension of the mechanism, not as a result established here.

8 Conclusion

Temporal coherence, variance preservation, and decorrelation form a useful representation-learning recipe that connects modern regularized self-supervision to SFA. TEREL turns that recipe into a concrete deep local rule: each nonlinear layer receives a temporal error, a gated neuron-local expansion signal, and a same-layer decorrelation signal. Canonical TEREL keeps short within-chunk temporal credit while cutting the error path across depth. TEREL-S additionally detaches time and has stream-length-independent state, although its dense lateral operator is quadratic in width.

Canonical TEREL reaches 97.30% on MNIST, within 1.04 points of data-presentation-matched BP and 1.95 points above the BatchNorm-calibrated random encoder. The final layer alone retains 97.14%, while temporally local TEREL-S reaches 96.29% and its tuned samplewise form learns a high-rank 95.14% validation representation. Matched Local SupCon reaches 96.98%, 0.32 points below TEREL across the paired seeds. One-factor controls further show that temporal coherence supplies the aligned signal, variance expansion prevents scale collapse, and decorrelation prevents redundancy collapse. These results substantiate the paper’s central intuition in a controlled label-ordered setting: temporal coherence, protected against collapse and redundancy, can train a two-hidden-layer representation with local credit assignment. Whether that advantage transfers to natural streams remains unresolved; the secondary result and explicit cost accounting mark the next experiments without diluting the controlled finding.

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A Gradient derivations

For valid detached pairs,

$$\frac{\partial}{\partial z_t} \frac{1}{N_I D} \sum_s I_s \|z_s - \text{sg}(z_{s-1})\|^2 = \frac{2I_t}{N_I D} (z_t - p_t). \quad (14)$$

Because g , m , A , and q_t are detached,

$$\frac{\partial L_V}{\partial z_t} = -\frac{2}{BD} g \odot (z_t - m), \quad (15)$$

$$\frac{\partial L_C}{\partial z_t} = \frac{1}{BD} A q_t. \quad (16)$$

Combining terms gives Eq. (8). Applying the chain rule first through ϕ and then through the minibatch map $N_{\eta, \mathcal{B}}$ gives the Jacobian-block form in Eq. (9); setting N to the identity recovers the displayed presynaptic outer-product special case. For undetached adjacent pairs, z_t appears as the current member of pair t and the previous member of pair $t + 1$; summing those derivatives gives Eq. (10). Finite-fixture regression tests check both signs and the presence of the future term.

For the direct covariance control, let Z_c stack centered rows and $C = \text{offdiag}(Z_c^\top Z_c / B)$. Since C is symmetric,

$$\frac{\partial}{\partial Z_c} \frac{\|C\|_F^2}{D} = \frac{4}{BD} Z_c C, \quad (17)$$

equivalently $4C(z_t - m)/(BD)$ in column-vector notation. The proxy replaces $4C(z_t - m)$ by the detached direction $A q_t$ and absorbs the positive scale into λ_C .

B Implementation-fidelity tests

The public suite checks:

- the negative variance gradient sign, variance gate, temporal detachment, undetached future derivative, lateral-proxy error bound, and direct covariance fixture;
- nonzero updates for every declared encoder layer and lateral state, with no gradient from a later layer to an earlier layer;
- fixed dynamic-state size, bounded streaming normalization, samplewise gradient release, and absence of retained autograd graphs;
- train-only BatchNorm calibration that changes running buffers while leaving every random-encoder weight, bias, and affine parameter unchanged;
- exact MNIST split roles, PAMAP2 subject roles, train-only normalization, temporal boundaries, and feature/boundary-only encoder batches;
- matched all-layer and last-layer probes, greedy layerwise scheduling, execution of every declared BP layer, and exact data-pass accounting;
- batch SFA constraints, boundary-aware derivatives, and numerical validation of the official-algorithm IncSFA port; and
- refusal of held-out evaluation after protocol, ledger, configuration, or tracked-source drift.

C Reproducibility record

The repository pins Python 3.12 and all direct dependencies in `pyproject.toml` and `uv.lock`. Runs enable deterministic PyTorch and cuDNN behavior and the prescribed cuBLAS workspace configuration. The anonymous supplement contains the exact final configurations, selection ledger, seeds, per-seed JSON records, raw confusion matrices, hardware/software record, source, generated tables, and checksums. Its `ARTIFACT_README.md` gives portable commands and the exact provenance identifiers; the manuscript retains the scientific protocol rather than that identifier ledger. Dataset files are not redistributed.

D Secondary natural-stream controls

The PAMAP2 encoder never receives activity labels. The chronological and shuffled TEREL rows differ only in order; the remaining rows locate their performance relative to untrained features, classical SFA, the exact in-batch covariance control, and end-to-end supervision. All methods are weak on subject 8, confirming that the split is dominated by cross-subject shift.

Table 10: Secondary PAMAP2 test macro-F1 across five seeds. Values are mean $\pm$ sample SD and percentile 95% bootstrap interval.

Method	Macro-F1
Chronological TEREL	0.307 ± 0.066 [0.260, 0.360]
Shuffled TEREL	0.312 ± 0.026 [0.290, 0.330]
Random features	0.322 ± 0.039 [0.296, 0.355]
Batch SFA	0.185 ± 0.002 [0.184, 0.186]
Incremental SFA	0.218 ± 0.015 [0.206, 0.229]
Direct covariance	0.243 ± 0.021 [0.226, 0.259]
Supervised BP	0.319 ± 0.030 [0.296, 0.341]

E Complete configurations and raw results

Table 11: Raw MNIST test accuracies (%) in seed order 1101, 1202, 1303, 1404, 1505.

Method	Raw values
TEREL (all layers)	97.30, 97.30, 97.21, 97.28, 97.41
TEREL (last layer)	97.14, 97.18, 97.28, 97.09, 97.01
TEREL-S (all layers)	96.35, 96.34, 96.32, 96.17, 96.26
Random, no normalization (all layers)	94.66, 95.27, 95.12, 95.33, 95.27
Backpropagation (all layers)	98.32, 98.32, 98.30, 98.28, 98.47

Table 12: Encoder-fit accounting averaged over five test seeds. Examples count sample presentations. MAC is the declared linear plus same-layer proxy, not a hardware energy measurement.

Method	Examples (M)	Opt. steps	Seconds	Peak MiB	MAC (G)
TEREL (all layers)	3.00	11760	107.6	76.7	3981.3
TEREL (last layer)	3.00	11760	107.7	76.7	3981.3
TEREL-S (all layers)	0.50	1960	27.4	77.7	1126.4
Random, no normalization (all layers)	0.00	0	0.0	0.0	0.0
Backpropagation (all layers)	3.00	11760	43.4	225.4	4815.4

Table 13: Training-state decomposition in MiB. Model parameters are those fitted during encoder training (the BP entry therefore includes its supervised head). Algorithm state is the TeReL lateral/statistics state after separating normalization buffers; optimizer and algorithm state are training-only. Inference encoder state retains hidden-layer parameters and normalization buffers but discards the BP head and TeReL state.

Method	Model params	Optimizer	Algorithm	Norm.	Inference encoder
TEREL (all layers)	2.040	4.080	1.262	0.006	2.046
TEREL (last layer)	2.040	4.080	1.262	0.006	2.046
TEREL-S (all layers)	2.034	4.068	1.262	0.000	2.034
Random, no normalization (all layers)	2.034	0.000	0.000	0.000	2.034
Backpropagation (all layers)	2.050	4.100	0.000	0.006	2.046

Table 14: Tuned samplewise streaming TeReL on the validation split (three seeds). The serial seed supplies time and memory accounting; concurrent repetition timings are excluded.

Accuracy (%)	Effective rank	Median variance	Encoder seconds
95.14 $\pm$ 0.04	112.8	0.378	1305.2

Table 15: Complete validation selection for Local SupCon (accuracy %, three seeds). Bold marks the predeclared winner.

Learning rate	Temperature	Accuracy
1e-04	0.1	96.49 $\pm$ 0.15
1e-04	0.2	95.60 $\pm$ 0.12
3e-04	0.1	96.63 $\pm$ 0.03
3e-04	0.2	95.98 $\pm$ 0.11
1e-03	0.1	96.90 $\pm$ 0.03
1e-03	0.2	96.20 $\pm$ 0.02

Table 16: Local SupCon five-seed evaluation values and resource state. Inference retains encoder parameters and batch-normalization buffers; optimizer state is training-only.

Quantity	Value
Test accuracy by seed (%)	96.97, 97.12, 96.99, 96.98, 96.83
Mean encoder seconds	82.1
Model parameters (MiB)	2.040
Optimizer state (MiB)	4.080
Normalization buffers (MiB)	0.006
Inference encoder (MiB)	2.046